

HELIX Toolbox — Spall Detection & Qualification

This note documents the spall-detection and did-not-spall (DNS) qualification logic in the HELIX Toolbox, including the improved pullback/recompression selection rule, using default configuration values and real 128 GS/s PDV free-surface velocity traces.

Pipeline overview

ALPSS → free-surface velocity (Savitzky–Golay smoothed, 6 ns) + uncertainty

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1. Uncertainty filter drop points with rel. uncertainty ≥ 1.0
2. Align $t = 0$ to ALPSS shock-arrival (signal start)
3. Window keep 0 – 100 ns after t_0
4. Gaussian smooth $\sigma = 1.5$ ns (spall-only; HEL uses ALPSS trace)
5. Plateau points $\geq 95\%$ of peak → P1, P2, plateau mean
6. Pullback → recomp. first prominent valley (P3) FOLLOWED BY a prominent recompression peak (P4) ← improved rule
7. Qualification 4 recompression gates as a final refinement

▼

VALID SPALL → σ_{spall} , strain rate, shock stress
or DNS → flagged with the failing reason

Default parameters

plateau_threshold	0.95	valley/plateau definition
spall_smoothing_sigma_ns	1.5	Gaussian smoothing (ns)
prominence_factor	0.01	min valley/peak depth / plateau vel.
peak_distance_ns	3.0	min spacing between features
min_recomp_ratio	0.015	rebound $\geq 1.5\%$ of pullback depth
min_recomp_velocity_ratio	1.015	P4 $\geq 1.5\%$ above valley
min_recomp_time_ns	2.5	P4 at least 2.5 ns after P3
spall window	0 – 100 ns	

Physics (applied only to VALID spall traces)

Spall strength $\sigma = \frac{1}{2} \cdot \rho \cdot c_b \cdot \Delta u_{\text{pullback}}$ ($\Delta u = \text{plateau} - P3$)
Strain rate $\varepsilon' = |\text{release slope}| / (2 \cdot c_b)$
Shock stress $\sigma = \rho \cdot U_s \cdot u_p$, $U_s = c + S \cdot u_p$, $u_p = \text{plateau}/2$

How the qualification logic works

Building the plateau (P1, P2)

The global velocity peak sets a 95% threshold. Every point at or above it is part of the plateau; their mean is the plateau velocity, and the earliest and latest such points become P1 and P2. Using the mean (not the single noisy peak) stabilises both the shock stress and the pullback measurement.

The spall signature: a pullback FOLLOWED BY a recompression

A spall leaves a two-sided fingerprint — the surface velocity drops into a pullback minimum (P3), then rebounds to a recompression peak (P4) as a wave reflects off the newly-formed internal spall surface. The improved rule requires BOTH to be present and prominent:

- P3 = a prominent local minimum after the plateau
($\text{depth} \geq \max(\text{prominence_factor} \times \text{plateau}, 1 \text{ m/s})$)
- P4 = the FIRST prominent local maximum after P3 ← local, not global
- take the first P3 (in time) that has such a P4; else DNS

The four recompression gates (final refinement)

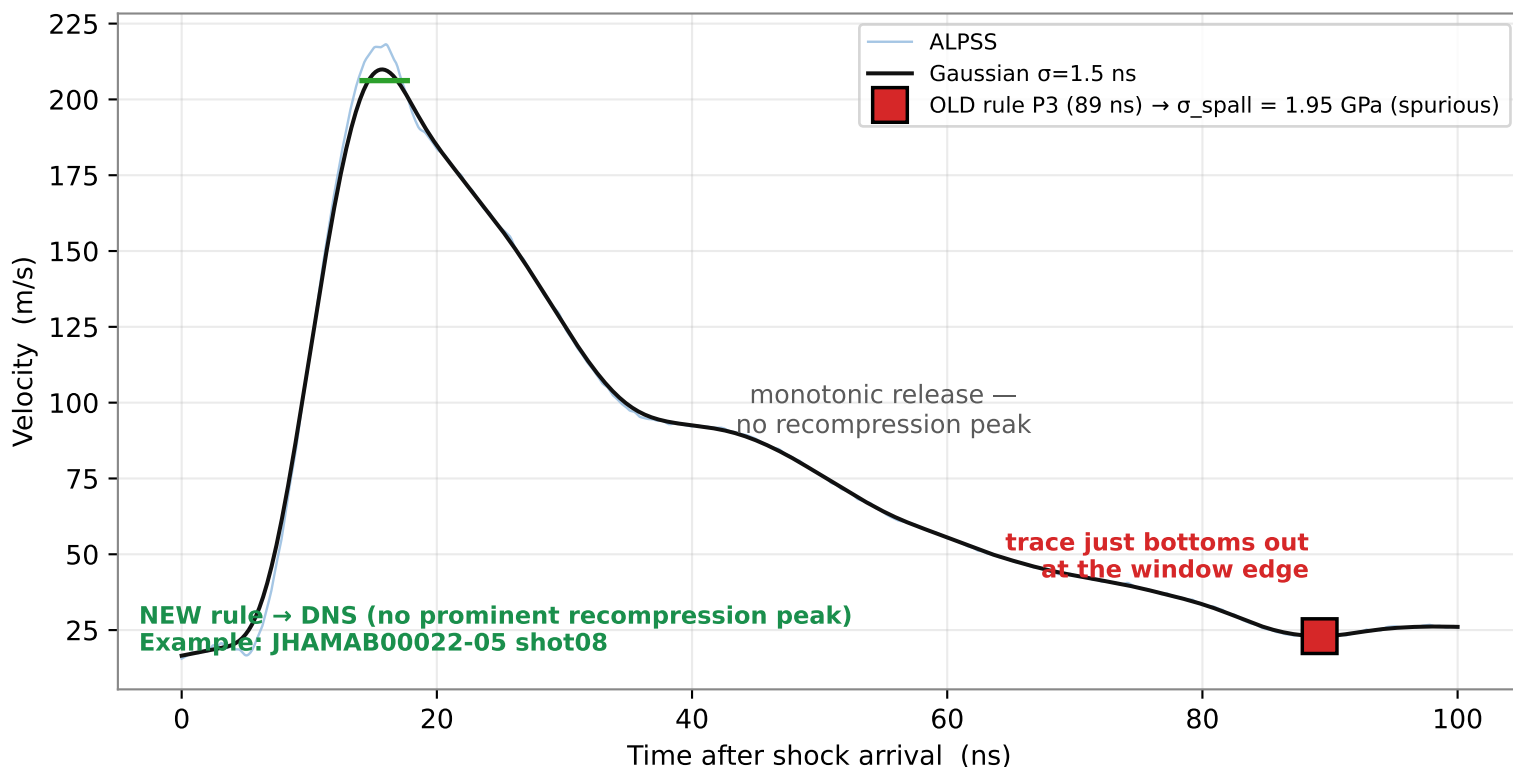
- | | | |
|----|---|-----------------------------------|
| 1. | $P4 > P3$ | there is some re-acceleration |
| 2. | $P4 \geq P3 \times 1.015$ | rebound vs the VALLEY velocity |
| 3. | $t(P4) - t(P3) \geq 2.5 \text{ ns}$ | rebound is sustained, not a spike |
| 4. | $(P4-P3) \geq 0.015 \times (\text{plateau}-P3)$ | rebound vs the pullback DEPTH |

Because a genuine recompression PEAK is now required for P3 to be accepted, the dominant DNS mode is simply 'no prominent recompression after the pullback' — i.e. the surface released without reforming a surface, which is physically exactly what did-not-spall means.

The next page explains why this rule replaced the previous one; the final two pages walk one VALID spall and one DNS trace through the logic.

Why the P3/P4 rule was improved

The previous rule took the FIRST valley clearing a fixed prominence bar as P3, then the GLOBAL maximum after it as P4. On a trace that shocks and then releases monotonically, that picks the near-baseline point where the trace bottoms out at the window edge — and fabricates a large 'pullback' with a tiny end-of-window uptick as the 'rebound':



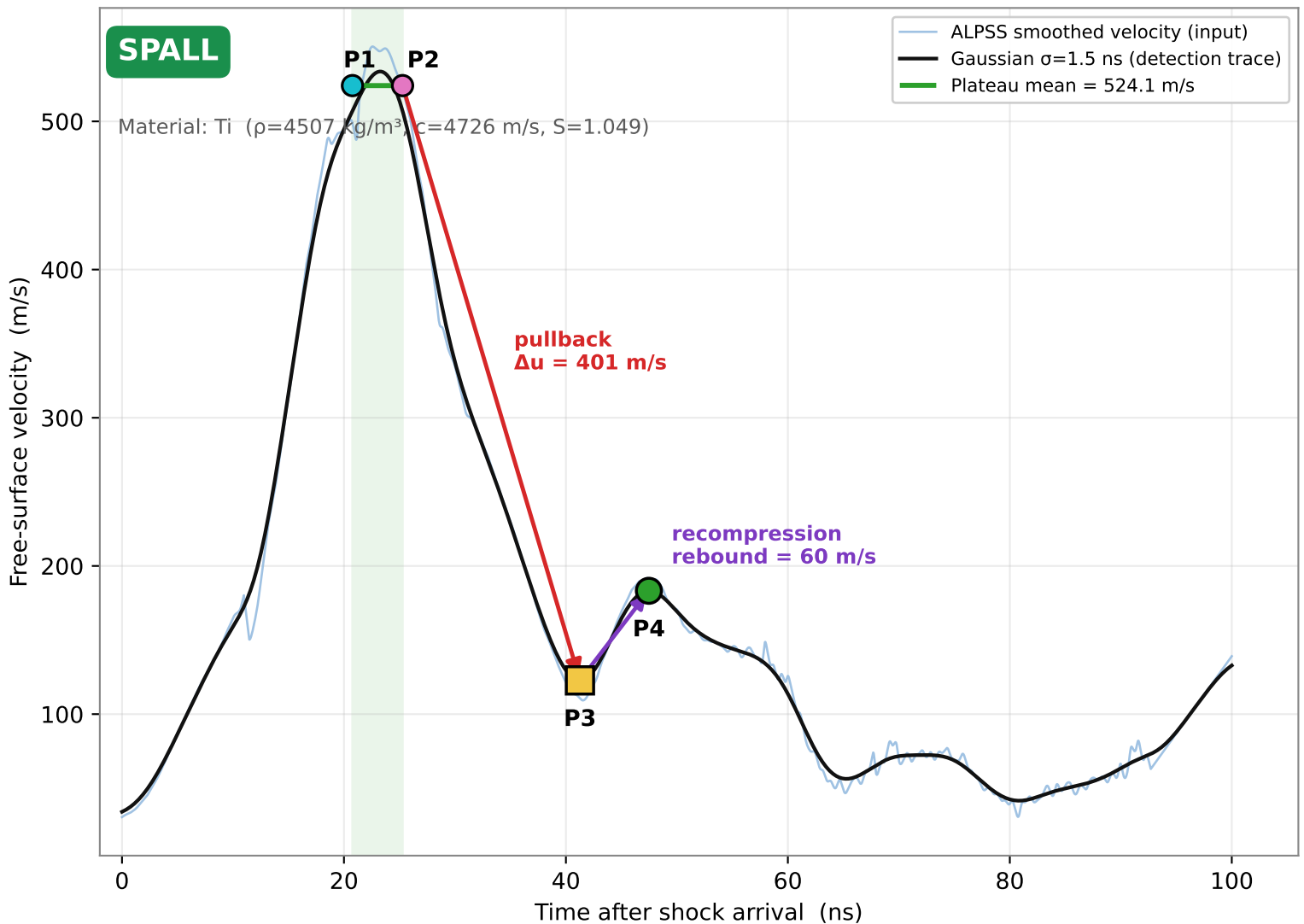
Validation across a 174-trace spall dataset (current defaults)

Previous rule	105 SPALL / 69 DNS
Improved rule	91 SPALL / 83 DNS
Verdicts changed	14 (all SPALL $\rightarrow$ DNS; 0 DNS $\rightarrow$ SPALL)
Flipped traces	old P3 was late (63–96 ns), spurious σ_{spall} up to 1.95 GPa fabricated from end-of-window decay
Retained spalls	91 traces — P3 location IDENTICAL (0.00 ns shift)
Ti reference trace	unchanged: $\sigma_{\text{spall}} = 4.275$ GPa under both rules

$\rightarrow$ The improved rule removes 14 false-positive spalls (monotonic releases mis-scored by the old end-of-window pick) and changes no legitimate detection.

Worked example 1 — VALID spall (Ti, CP-Grade)

JHAMAL00016-004 shot20 — identical under the previous and improved rules



Detection — pullback → recompression signature

Prominent pullback valley (P3) + recompression peak (P4)

yes

Recompression gates (final refinement)

$P4 > P3$ (any re-acceleration)

183.2 > 122.8 ✓

$P4 \geq P3 \times 1.015$ (rebound vs valley)

183.2 vs 124.6 ✓

$t(P4) - t(P3) \geq 2.5 \text{ ns}$ (duration)

6.21 ns ✓

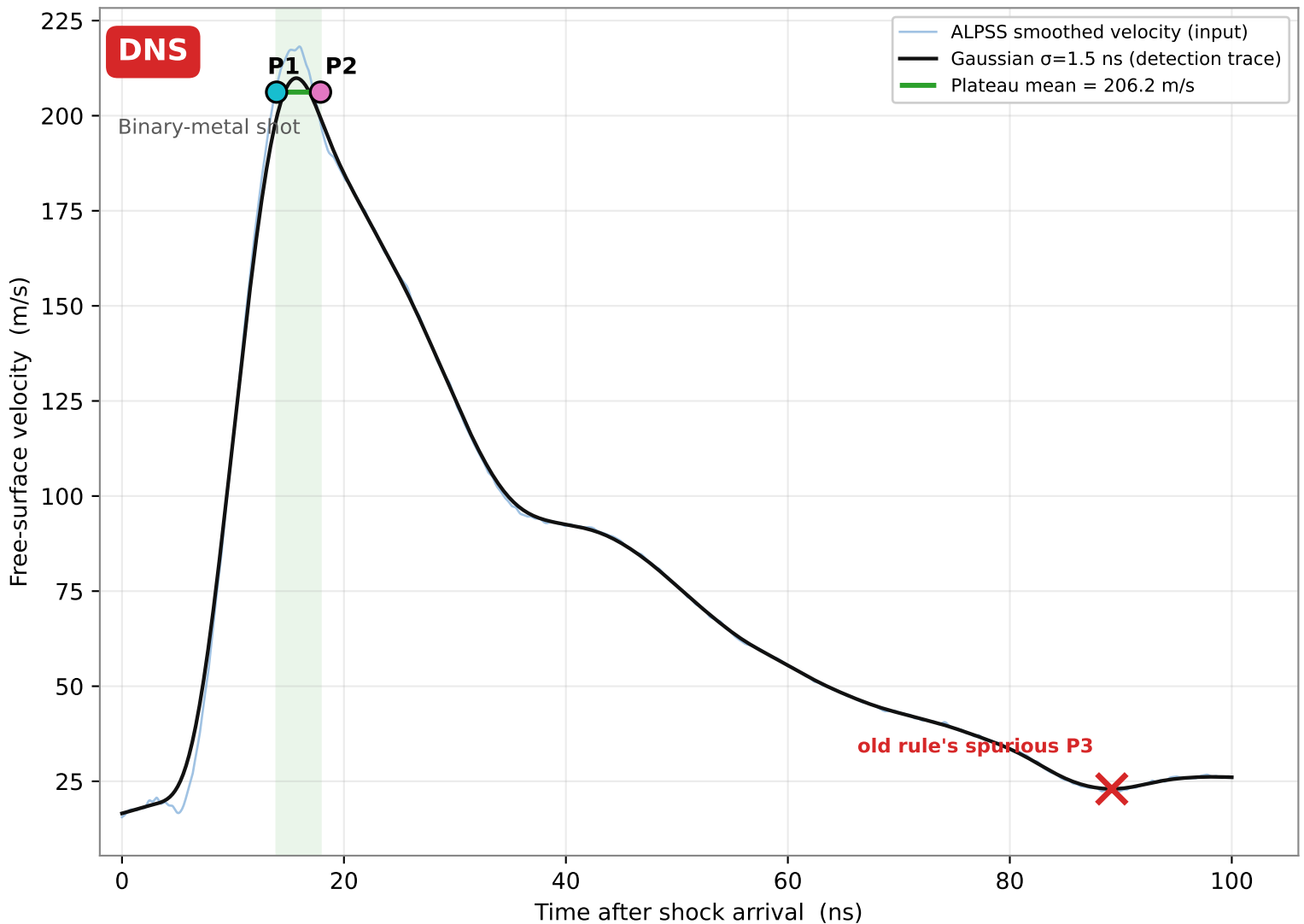
$\text{rebound} \geq 0.015 \times \text{pullback}$ (vs depth)

60.5 vs 6.0 ✓

→ **VALID SPALL** $\sigma_{\text{spall}} = 4.275 \text{ GPa}$, $\dot{\epsilon} = 2.66\text{e}+06 \text{ s}^{-1}$, $\sigma_{\text{shock}} = 5.907 \text{ GPa}$

Worked example 2 — DNS (no recompression)

JHAMAB00022-05 shot08 — shocks, then releases with no recompression peak



Detection — pullback → recompression signature

Prominent pullback valley (P3) + recompression peak (P4)

NO

→ **DNS: No valley with a prominent recompression peak**